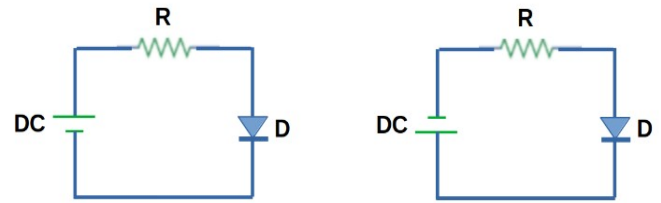


Tutorial series N°1

Problem 1 :

Calculate the current I_D for the following circuits:

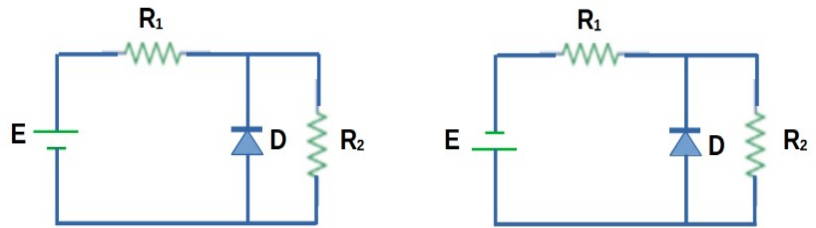
$$DC=5V ; V_D=0.6V \text{ and } R=1k\Omega$$



Problem 2:

Consider the following circuits: $V_D=0.7V$ and $R_1=R_2=1k\Omega$

- For $E=10V$, calculate the voltage across R_2 .
- The DC voltage source E is replaced by an AC voltage source:
 $e(t)=10\sin(2\pi t/T)$, with $T=20ms$.
- Determine the voltage across R_2 .

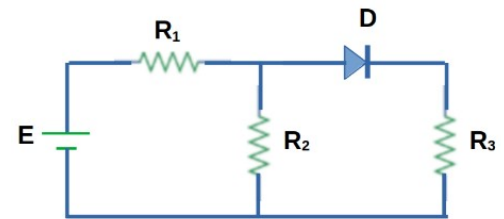


Problem 3:

For the following circuit:

$$R_1=2k\Omega, \quad R_2=1k\Omega, \quad R_3=2k\Omega$$

- Calculate the current and the voltage across resistor R_3 in the following cases: $E=5V$ and $E=1.5V$ ($V_D=0.6V$).

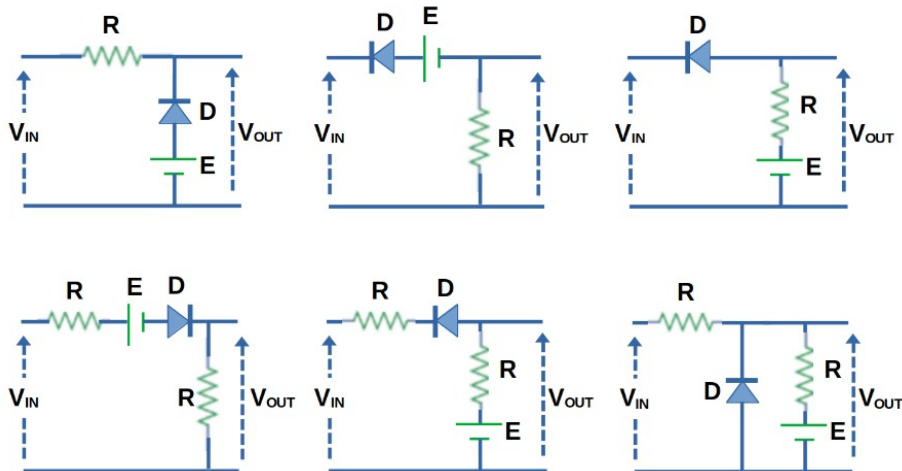


Problem 4:

For each of the given circuits, the diode is considered ideal.

$$\text{Given: } V_{\text{Input}}(t)=12\sin(\omega t) \text{ and } E=3V.$$

- Plot the transfer characteristic $V_{\text{Out}}(t)=f[V_{\text{Input}}(t)]$.
- Plot $V_{\text{Out}}(t)$.

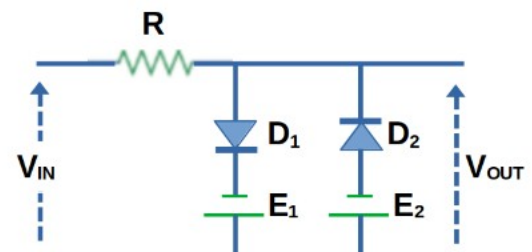


Problem 5:

Consider the circuit opposite, D_1 and D_2 are assumed to be ideal

$$\text{Given: } V_{\text{Input}}(t)=15\sin(\omega t), \quad E_1=5V \text{ and } E_2=10V.$$

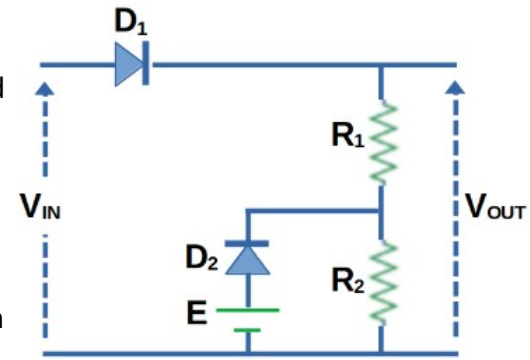
- Plot the transfer characteristic $V_{\text{Out}}(t)=f[V_{\text{Input}}(t)]$.
- Plot $V_{\text{Out}}(t)$.



Problem 6:

Consider the circuit opposite with: $E=10\text{ V}$ and $R_1=R_2=1\text{ k}\Omega$. D_1 and D_2 are assumed to be ideal.

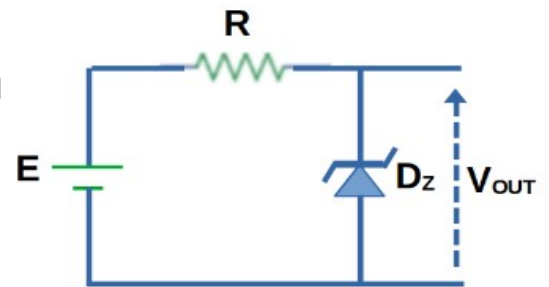
1. Plot the transfer characteristic of the circuit for $0 < V_{\text{Input}} < 15\text{ V}$.
2. Plot the representative curve of the current flowing through the resistance R_1 for $0 < V_{\text{Input}} < 15\text{ V}$.



Problem7:

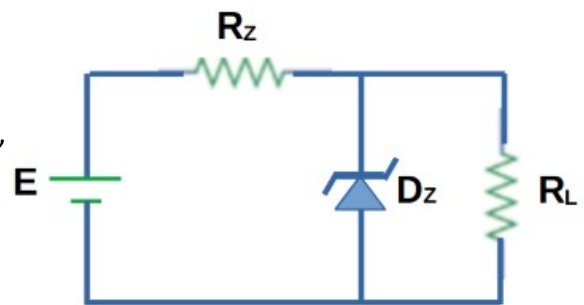
Consider the circuit in the figure on the right with: $E=50\text{ V}$ and $R=4\text{ k}\Omega$. Find V_{OUT} when:

1. The diode is ideal.
2. The diode is characterized by $V_{Z0}=25\text{ V}$ and $R_Z=10\text{ }\Omega$.



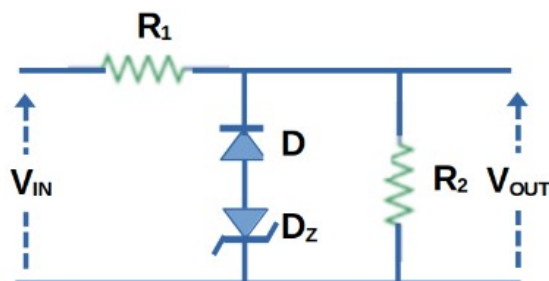
Problem8:

Find the current through the Zener diode given: $E=15\text{ V}$, $R_Z=1\text{ k}\Omega$, $V_Z=9\text{ V}$, and $R_L=2\text{ k}\Omega$



Problem9:

Consider the circuit shown below. The input voltage is: $V_{\text{INPUT}}(t)=36\sin(\omega t)$. The circuit parameters are: $R_1=150\text{ }\Omega$; $R_2=100\text{ }\Omega$; Diode ($V_0=0.7\text{ V}$, $R_D=0\text{ }\Omega$); and for Zener diode D_Z ($V_Z=10\text{ V}$; $R_Z=R_D=0\text{ }\Omega$ and $V_0=0\text{ V}$)



- Draw the output voltage